SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA1004	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	R.Nandhini	**Reg. No.:**	192524355
Date:	9/10/2026	**Duration:**	60 minutes

Code of Conduct

I R.Nandhini / 192524355 *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _____

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. **Understand the Problem Statement**
   - Explain the problem in your own words.
   - Identify the objectives of the proposed system.
   - State the scope of the problem.
2. **Identify Key Stakeholders and Challenges**
   - Identify all the stakeholders involved in the system.
   - Explain the roles and responsibilities of each stakeholder.
   - Discuss the major challenges and issues faced by the stakeholders.
3. **Analyze the Current Approach**
   - Describe the existing system or current process.
   - Identify its strengths and limitations.
   - Analyze the problems or gaps that need to be addressed.
4. **Propose a Suitable Solution Using Software Engineering Principles**
   - Design a software-based solution for the given problem.
   - Describe the major functional modules of the proposed system.
   - Identify the functional and non-functional requirements.
   - Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.
5. **Justify the Chosen Methodology/Framework**

- Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
- Justify why the selected methodology/framework is suitable for the proposed system.
- Explain how it supports successful project development and maintenance.

6. **Discuss Expected Outcomes and Limitations**
   - Describe the expected benefits of the proposed solution.
   - Explain how the solution addresses the identified challenges.
   - Discuss potential risks, implementation challenges, and limitations.
   - Suggest possible future enhancements.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework	Selects the most suitable	Suitable methodology	Methodology selected but	Outcomes, limitations, or

Justification (10%)	SDLC/model/framework and provides strong technical justification.	selected with adequate justification.	justification is weak.	future enhancements are missing; report lacks organization and clarity.

Criteria	**Mark**
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25

INTELLIGENT VEHICLE MAINTENANCE MANAGEMENT SYSTEM

Project Management using a Kanban Board

SUBMITTED BY: R.NANDHINI

REGISTER NUMBER: 192524355

DEPARTMENT : B-TECH AIDS

SUBMITTED TO: DR.K.ANITADAVAMANI

COLLEGE: SIMATS UNIVERSITY

1. AIM

To create and organize a Jira project for the Intelligent Vehicle Maintenance Management System by adding development tasks and tracking their progress using a Kanban board.

2. INTRODUCTION

Organizations that operate cars, buses, trucks, vans, and other vehicles must maintain them regularly. Servicing helps vehicles remain safe, dependable, and ready for use. When service dates are missed or faults are not recorded, vehicles may break down, repairs may become expensive, and daily operations may be delayed.

The Intelligent Vehicle Maintenance Management System is a proposed system that organizes vehicle details, service histories, maintenance schedules, faults, reminders,

spare-parts usage, repair costs, and vehicle utilization. Jira is used in this practical exercise to plan and monitor the tasks needed to develop the system.

3. PROBLEM STATEMENT

Fleet operators may manage many vehicles, each with different service dates, usage levels, faults, and repair requirements. If these details are kept manually in notebooks or separate files, information may be missed or become difficult to find. A vehicle may not receive servicing on time, a reported fault may remain unresolved, or repair expenses may not be tracked properly.

The proposed Intelligent Vehicle Maintenance Management System aims to keep vehicle maintenance information organized and help staff plan service activities. It is intended to support timely maintenance reminders, fault tracking, repair-cost monitoring, and fleet maintenance reporting.

4. NEED FOR THE SYSTEM

- ✓ To keep vehicle information in an organized place.
- ✓ To avoid missing scheduled maintenance.
- ✓ To record vehicle faults and follow their repair status.
- ✓ To maintain service and repair history for each vehicle.
- ✓ To track spare parts and repair expenses.
- ✓ To understand vehicle usage and plan servicing more effectively.

5. AIM AND SCOPE OF THE JIRA EXERCISE

The practical work focuses on using Jira to organize project activities. A Jira space named “Intelligent Vehicle Maintenance Management System” has been created with a Kanban board. Work items are represented as cards and arranged in To Do, In Progress, and Done

columns. This board tracks the development work; it is not the completed vehicle-maintenance software itself.

6. OBJECTIVES

- ✓ Maintain vehicle records such as registration number, type, model, and current status.
- ✓ Track service history, including service date, work performed, and repair details.
- ✓ Schedule preventive maintenance based on time or vehicle usage.
- ✓ Record faults, their severity, and whether the repair is pending or completed.
- ✓ Generate reminders for upcoming or overdue maintenance.
- ✓ Track spare parts used during repair and record repair costs.
- ✓ Monitor vehicle utilization, such as distance travelled or operating hours.
- ✓ Generate reports that summarize maintenance activities and fleet expenses.

7. PROPOSED SYSTEM MODULES

Vehicle Records: Stores basic details for each vehicle, such as vehicle number, type, model, and status.

Service History: Maintains previous service dates, repairs, and notes about work completed.

Preventive Maintenance: Helps plan routine service before a fault becomes serious.

Fault Management: Records vehicle faults and follows them until they are resolved.

Maintenance Reminders: Identifies upcoming or overdue service based on date or usage.

Spare Parts and Repair Costs: Records parts used and expenses for each repair.

Vehicle Utilization: Tracks distance or operating time to support service planning.

Fleet Reports: Summarizes vehicle servicing, faults, costs, and maintenance status.

8. PROPOSED WORKFLOW

The proposed application can follow a simple workflow. First, the staff member adds the vehicle details. The service history is then recorded whenever maintenance is performed. The system checks service dates or usage information to identify upcoming maintenance. If a fault is reported, it is recorded and assigned for repair. The repair details, spare parts, and costs are updated after the work is completed. Finally, fleet reports can be reviewed by the manager.

Add vehicle details.

- ✓ Record service and usage information.
- ✓ Check the next maintenance due date.
- ✓ Record faults and repair requirements.
- ✓ Send or display maintenance reminders.
- ✓ Update spare-parts usage and repair costs.
- ✓ Review fleet maintenance reports.

9. EXPECTED BENEFITS

- ✓ More organized vehicle records and service histories.
- ✓ Better planning of preventive maintenance.
- ✓ Faster identification and tracking of faults.
- ✓ Improved awareness of repair costs and spare-parts usage.
- ✓ Reduced chance of unexpected breakdowns.
- ✓ Improved vehicle safety and operational reliability.

10. ROLE OF JIRA

Jira helps the project team organize development tasks, see their current status, and understand what remains to be done. The Kanban board provides a visual workflow. It does not automatically monitor vehicles or generate real maintenance alerts unless such features are implemented in the actual application.

11. JIRA IMPLEMENTATION: STEP-BY-STEP

Open Jira: Sign in to Jira and open the project or space area.

Create a space: Create a space named Intelligent Vehicle Maintenance Management System.

Select a template: Choose the Kanban template to use a visual task board.

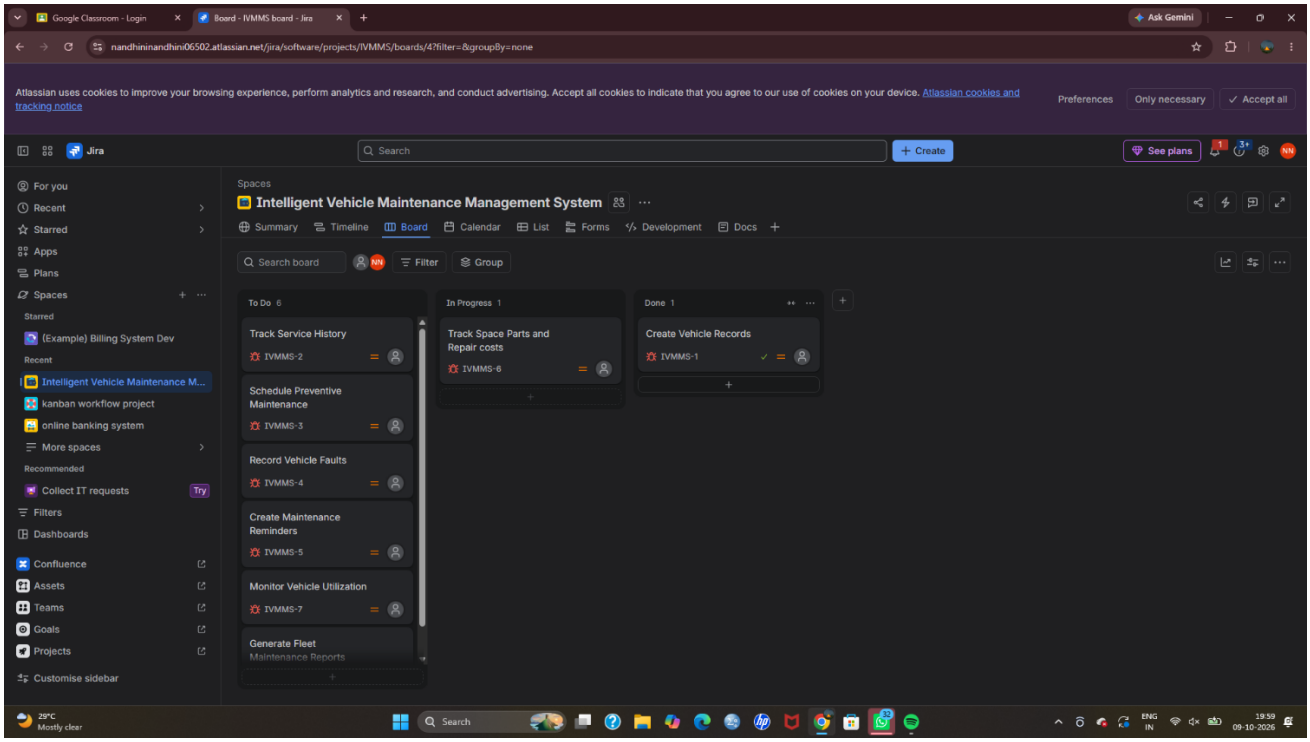
Open the board: Open the Board tab and check the To Do, In Progress, and Done columns.

Add tasks: Create work items for the major objectives of the proposed system.

Arrange tasks: Place each work item in the column that matches its current progress.

Update progress: Move a card to In Progress when work begins and to Done only after completion.

Review and capture: Check the board and take a screenshot as evidence for the practical report.



12. KANBAN BOARD COLUMNS

To Do: Tasks that have not started.

In Progress: Tasks currently being worked on.

Done: Tasks that have been completed.